

MANPREET SINGH

@ manpreetchahal0786@gmail.com

+91 8054651374

Gurgaon, Haryana

in [LinkedIn](#)

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SUMMARY

Data Engineer with experience in building **real-time data pipelines**, **CDC systems**, and **ETL workflows** using Kafka, Debezium, and NiFi. Skilled in distributed systems, scalable data architecture, SQL optimization, Python, and cloud infrastructure, with expertise in reliable and high-performance distributed data processing.

EDUCATION

B. Tech. Computer Engineering | Thapar University, Patiala

Aug 2020 – May 2024

- CGPA: 8.4 | Relevant Coursework: **Data Structures, Algorithms, Machine Learning, Data Science, Database Management Systems, Web Development**

WORK EXPERIENCE

Data Engineer | RapidShyp

Gurgaon

Jun 2025 – Present

- Designed and maintained scalable **real-time data pipelines** using **Debezium, Apache Kafka, and Apache NiFi** to capture and process **CDC events from MongoDB to PostgreSQL (OLTP → OLAP flow)**.
- Implemented **ETL/ELT data pipelines** with schema transformation, validation, and error handling, improving **data reliability and consistency**.
- Optimized **data ingestion and staging pipelines**, reducing data latency and improving throughput for **high-volume logistics transactions**, processing over **2M events daily**.
- Led **AWS to GCP migration** of data infrastructure, including cron jobs and pipeline services, ensuring **minimal downtime during the production cutover**.
- Developed **Python automation scripts** for data reconciliation, monitoring, observability, and alerting, reducing manual operational effort by **80%**.
- Worked on AI-enabled data workflows and data pipelines, integrating **LLMs and RAG architectures**.

Associate Application Engineer | RapidShyp

Gurgaon

Jun 2024 – Jun 2025

- Resolved **300+ production issues** across **20+ microservices**, debugging distributed systems and APIs while maintaining **99.9% uptime**.
- Performed **root cause analysis** using **SQL, Kafka, MongoDB, and log analysis**, reducing issue resolution time by **40%**.
- Developed **data extraction and analysis utilities** to support debugging and operational insights.
- Contributed to **monitoring, alerting, and system reliability improvements**, reducing incident recurrence by **35%**.

TECHNICAL SKILLS

Data Engineering

Apache Kafka, Debezium, Apache NiFi, ETL/ELT, CDC (Change Data Capture), Data Pipelines, Data Modeling, Data Warehousing, Analytics, Schema Design, Data Validation, Batch & Real-time Processing, Event-driven Architecture

Databases

MongoDB, PostgreSQL, MySQL, ClickHouse, Snowflake

Languages

Python, SQL, C++, JavaScript, Shell Scripting, Node.js, REST APIs

Cloud & Tools

AWS, GCP, Apache Airflow (Workflow Orchestration), PySpark, Linux (Ubuntu), Git, GitHub, Jira, Superset, Tableau, **LLMs, RAG, LangChain, LangGraph, MCP, Prompt Engineering, Generative AI Workflows**

ACHIEVEMENTS & CERTIFICATIONS

- Solved **500+ Data Structures & Algorithms problems** across major platforms.
- Certified **Coursera Data Science Specialization**. [\[Certificate\]](#)
- Earned the **AWS Machine Learning Foundations** badge. [\[Certificate\]](#)
- Published research paper on “**Li-Fi Based Toll Collection System**” at **ICICC-2025**. [\[Certificate\]](#) [\[Publication\]](#)
- Secured **93.84%** in ICSE 10th and **90.6%** in CBSE 12th.